As part of a team, I helped develop Crazy Math, a timed math game running on a DE-10 development board. The objective was to earn ten points by correctly solving equations before the timer ran out. Each correct answer awarded one point and reset the timer, while an incorrect answer cost the player a life without resetting the timer. The project required us to develop a playable game that used the board's physical inputs and displays.

My responsibilities included programming the main game loop, equation generator, and wall timer, as well as integrating the code with the DE-10 hardware. The game was written in C. Players entered numerical answers using the board's switches, pressed Button 0 to submit an answer, and used Button 1 to reset the game. The score appeared on the seven-segment display, while the equations, remaining lives, and the wall timer appeared on the main display.

One of the biggest challenges was working with the display. I had little prior experience with display programming, and getting the visuals to stay synchronized with the game's internal state required lots of troubleshooting. I also encountered a problem with the moving walls: when their edges met in the middle, they continued moving instead of resetting. To fix this, I implemented a check that detected when the edges were next to or overlapping each other, allowing the program to reset them.

The finished game was completed and playable. This project gave me experience working in an area I had not worked with much before, particularly display programming and integrating software with physical hardware. It also gave me an opportunity to troubleshoot problems involving both the game's logic and its visual output.